SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

For Consult Kenya: Visual-First Professional Advisory & Authority Network

Conforming to ISO/IEC/IEEE 29148:2018 International Standard

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) establishes the definitive, complete, and verifiable functional and non-functional requirements for the **Consult Kenya** platform. It serves as the authoritative technical baseline for software architects, backend developers, frontend engineers, quality assurance auditors, escrow mediators, and compliance officers developing the system.

1.2 Scope of the System

Consult Kenya is a hybrid professional advisory ecosystem and authority-driven knowledge network designed to bridge the structural divide between traditional, visually sterile professional networks (e.g., LinkedIn) and high-engagement, visual-first consumer platforms (e.g., Instagram).

The platform transforms dry professional resumes and text articles into **swipeable visual case studies, architecture schemas, and micro-video walkthroughs**, backed by **verifiable third-party proof of execution** (statutory licenses, GitHub commits, enterprise client sign-offs). The system closes the loop from passive consumption to commercial transaction through an **embedded zero-redirect booking engine**, **M-Pesa STK push and multi-currency card payment rails**, and an **enterprise milestone escrow vault** designed for high-value B2B advisory contracts.

1.3 Definitions, Acronyms, and Abbreviations

- **ACID**: Atomicity, Consistency, Isolation, Durability (database transaction properties).
- **ANN**: Approximate Nearest Neighbor (high-dimensional vector search).

- **CAS**: Creator Authority Score (the platform's proprietary multi-factor authority ranking metric, bounded from 0.00 to 10.00).
- **CBK**: Central Bank of Kenya.
- **CIArb**: Chartered Institute of Arbitrators (Kenya Branch).
- **Daraja**: Safaricom's M-Pesa API integration gateway.
- **DPA 2019**: Kenya Data Protection Act, No. 24 of 2019.
- **EBK**: Engineers Board of Kenya.
- **eTIMS**: Electronic Tax Invoice Management System (Kenya Revenue Authority statutory mandate).
- **ICPAK**: Institute of Certified Public Accountants of Kenya.
- **IPRS**: Integrated Population Registration System (Kenyan National Identity Gateway).
- **LSK**: Law Society of Kenya.
- **PWA**: Progressive Web App.
- **RLS**: Row-Level Security (PostgreSQL / Supabase database multi-tenant isolation mechanism).
- **SOW**: Statement of Work.
- **STK**: SIM Toolkit (automated M-Pesa push prompt dispatched directly to client handsets).
- **WebRTC**: Web Real-Time Communication (peer-to-peer browser video/audio streaming).
- **WHT**: Withholding Tax (statutory 5% professional service tax in Kenya).

1.4 References

4. **ISO/IEC/IEEE 29148:2018**: *Systems and software engineering — Life cycle processes — Requirements engineering*.
5. **ISO/IEC 25010:2011**: *Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — System and software quality models*.
6. **Jackson, M., & Zave, P. (1997)**: *Domain Descriptions and Requirements Engineering*, Requirements Engineering, 2(1), 14-21.

4. **Kenya Data Protection Act, 2019**: Republic of Kenya, Acts No. 24 of 2019.

5. **Tax Procedures (Electronic Tax Invoice) Regulations, 2024**: Kenya Revenue Authority (KRA).

1.5 Document Overview

Section 2 provides the high-level system perspective, personas, operational constraints, and lifecycle evolution roadmap. Section 3 details all **22 Core System Features** organized across 6 orthogonal domains. Section 4 specifies external interfaces. Section 5 formalizes the 16 non-functional quality requirements with measurable fit criteria. Section 6 provides the verification and traceability matrix.

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2. OVERALL DESCRIPTION

2.1 Product Perspective & Context

Consult Kenya operates as a multi-sided digital marketplace and professional advisory network. The platform solves the classic marketplace "cold-start" problem through a **3-Phase Evolution Roadmap** (`CARD-002`):

- **Phase 1 (The Anchor Agency):** The system launches as a proprietary flagship consulting and systems architecture agency. The founding agency generates 100% of the initial visual case studies, establishing the content aesthetic and proving monetization before opening onboarding.
- **Phase 2 (The Curated Cohort):** An invite-only beta unlocking creator tools for 25 hand-picked specialist peers (software architects, corporate advocates, tax consultants) using cryptographic invitation codes.
- **Phase 3 (The Open Ecosystem):** Full self-service onboarding across Kenya and East Africa with automated KYC credential vetting and multi-tenant SaaS profile tiers.

```
flowchart LR
    P1["Phase 1: Anchor Agency<br>(100% Proprietary Content)"] --> P2["Phase 2:
Curated Cohort<br>(Invite-Only Beta of 25 Specialists)"]
    P2 --> P3["Phase 3: Open Ecosystem<br>(Vetted Multi-Tenant Marketplace)"]
```

2.2 Product Functions & High-Level Architecture

The system integrates six major functional subsystems:

7. **Dual-Mode Consumption Subsystem**: Provides two contextually separated modes of interaction:

- *Vertical Stream (Home)*: High-engagement, full-screen vertical discovery stream for rapid passive learning.
- *Bento Grid (Profile & Explore)*: High-density modular grid layout for 5-second competence due diligence.

8. **Creation Studio & Narrative Template Engine**: 4-step wizard (*Format $\to$ Template $\to$ Evidence $\to$ Conversion Anchor*) with 300-char slide limits and split-screen preview.
9. **Trust & Credential Verification Subsystem**: Automated API hooks into Kenyan National ID (IPRS), KRA PIN, and professional statutory boards (LSK, ICPAK, EBK).

4. **Native Booking & Scheduling Subsystem**: Zero-redirect slide-up calendar sheet with distributed 10-minute mutex slot locking and pre-intake questionnaires.

5. **Fintech Payment Subsystem**: Dual-rail checkout supporting instant Safaricom M-Pesa STK Push, manual Paybill fallbacks, and multi-currency international card rails.

6. **Milestone Escrow & Mediation Subsystem**: Dedicated trust ledger securing enterprise project deposits, enforcing 72-hour review periods, SHA-256 asset delivery proof, 7-day auto-release fallbacks, and binding dispute mediation.

2.3 User Classes, Characteristics, and Personas

The system serves four primary stakeholder classes:

10. **Client / Enterprise Buyer (Persona: David, Enterprise CTO & Corporate Procurement)**:

- *Profile:* Technical decision-makers and business founders seeking specialized external expertise.
- *Needs:* Fast due diligence, visual proof of past work, zero-friction booking, and financial protection via milestone escrow.

11. **Consultant / Specialist Creator (Persona: Leah, Lead Systems Architect & High Court Advocate)**:

- *Profile:* Top-tier vetted Kenyan professionals offering advisory, legal, architectural, or financial engineering services.
- *Needs:* Streamlined case study authoring, protection against client non-payment, instant mobile payouts, and algorithmic distribution based on merit rather than superficial likes.

12. **Platform Administrator & Moderator**:

- *Profile:* Internal operations team managing platform governance.
- *Needs:* Phase rollout gating, automated spam/fluff filtering, and emergency 4-hour SLA enterprise NDA breach takedowns.

4. **Escrow Mediator & Compliance Officer**:

- *Profile:* Certified dispute arbitrators and finance compliance officers.
- *Needs:* Chronological contract audit dashboards, binding dispute split execution, and KRA eTIMS statutory tax reporting.

2.4 Operating Environment & Technical Dependencies

- **Client Frontend**: Progressive Web App (PWA) built with modern web technologies, responsive from 360px mobile viewports to 4K desktop displays.
- **Database & Auth Layer**: PostgreSQL with strict Row-Level Security (RLS) policies and multi-tenant schema partitioning.
- **Vector Indexing Engine**: In-memory dense semantic vector index (128-dimensional embedding space) supporting sub-50ms cosine similarity queries.
- **Video / Media Streaming**: WebRTC peer-to-peer audio/video with DTLS-SRTP encryption and distributed TURN relay fallbacks.
- **External API Gateways**:
- Safaricom Daraja API (C2B, B2C, STK Push).
- Paystack & Flutterwave (Multi-currency cards).

- Google Calendar & Microsoft Outlook Graph APIs.
- KRA eTIMS Fiscal Electronic Invoicing Gateway.

2.5 Design and Implementation Constraints

- **DC-01 (Statutory Compliance)**: Must strictly comply with Kenya Data Protection Act (DPA 2019) regarding local data residency for PII.
- **DC-02 (Fiscal Invoice Mandate)**: Every transaction fee deducted by the platform must immediately generate a KRA eTIMS QR-coded tax receipt.
- **DC-03 (Anti-Clutter Visual Rule)**: Case study slides in Creation Studio are hard-constrained to ≤ 300 characters to maintain scannability.
- **DC-04 (Zero Off-Platform Redirection)**: Consultation booking must occur natively inside an in-app slide-up sheet without redirecting to third-party scheduling tools.

2.6 Assumptions and Dependencies

- **AS-01**: Mobile telecom operators in Kenya maintain operational STK push gateway connectivity with average response times under 5 seconds.
- **AS-02**: Statutory professional boards (LSK, ICPAK, EBK) maintain accessible digital registries or verify active practitioners via designated API endpoints.
- **AS-03**: Clients reviewing milestone deliverables possess modern web browsers capable of rendering PDF documents and vector architecture diagrams.

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3. SPECIFIC REQUIREMENTS & SYSTEM FEATURES (ALL 22 CORE USE CASES)

```
mindmap
  root((Consult Kenya Core Features))
    Domain A: Content & Discovery
      UC-01: Dual-Mode Feeds
      UC-02: Bento Grid Profile
      UC-03: Topic Hubs Search
      UC-04: Bookmarks
    Domain B: Authoring & Trust
      UC-05: Creation Studio
      UC-06: Evidence Tagging
      UC-07: Conversion Anchors
      UC-08: Credential Verification
    Domain C: Scheduling & Advisory
      UC-09: Native Booking Sheet
      UC-10: M-Pesa STK Push
      UC-11: Card Payments
      UC-12: WebRTC Live Call
    Domain D: Milestone Escrow
      UC-13: Fund Escrow Contract
```

```
  UC-14: Deliverable Proof SHA-256
  UC-15: Approve & Payout
  UC-16: Escrow Dispute
Domain E: Algorithmic Brain
  UC-17: Calculate CAS Score
  UC-18: Two-Pass Feed AI
Domain F: Governance & Tax
  UC-19: Phased Rollout
  UC-20: Content Moderation
  UC-21: Adjudicate Dispute
  UC-22: KRA eTIMS Tax Audit
```

3.1 Domain A: Content Consumption & Discovery

UC-01: Discover Content via Dual-Mode Feeds

- **Description:** The system provides an immersive vertical stream on the Home tab delivering bite-sized visual case studies and micro-videos.
- **Inputs:** Vertical swipe gestures, horizontal slide drag gestures, network downlink telemetry.
- **Outputs:** Rendered 1080p WebP slides, muted auto-playing video streams, slide progress scrubbers.
- **Functional Requirements:**
- **FR-01.1:** The system shall render vertical feed cards with a transition latency $\le 200\text{ms}$ (P_{95}).
- **FR-01.2:** When a user drags horizontally on a multi-slide carousel by $>15\text{px}$, the system shall lock vertical feed scrolling until the gesture terminates.
- **FR-01.3:** The system shall automatically downgrade video playback to 70% quality WebP image slides when detected network bandwidth falls below 1 Mbps.

UC-02: Inspect Consultant Portfolio (Bento Grid)

- **Description:** When a user opens a consultant's profile, the system presents a high-density Bento Grid showcasing their verified authority baseline in under 5 seconds.
- **Inputs:** Avatar tap, profile URL query, contextual mode switch toggle.
- **Outputs:** Multi-tile Bento Grid layout (Flagship deliverables, statutory verification seal, star ratings, hourly rate, sticky booking button).
- **Functional Requirements:**
- **FR-02.1:** The system shall render all primary authority indicators above the fold without requiring user scrolling.
- **FR-02.2:** The system shall support a sub-toggle switching between "Bento Grid" and "Chronological Feed" without triggering a page reload.
- **FR-02.3:** Newly registered consultants with zero posts shall display guided cold-start credential badges and onboarding banners.

UC-03: Search & Filter Topic Hubs

- **Description:** The system provides semantic natural language search and multi-facet filtering across industry topic hubs.
- **Inputs:** Search query string, filter chips (`#SoftwareArchitecture`, `#CorporateLaw`, `#TaxStrategy`, minimum rating, verified tier).
- **Outputs:** Ranked candidate cards, categorized topic hubs, nearest parent category suggestions.
- **Functional Requirements:**
- **FR-03.1:** The system shall convert query strings into 128-dimensional dense vectors and execute cosine similarity matching within $\le 50\text{ms}$.
- **FR-03.2:** The system shall support multi-facet filtering by discipline, minimum star rating, and verification tier.
- **FR-03.3:** When zero exact matches occur, the system shall suggest nearest semantic topics and return top-rated consultants in the parent discipline.

UC-04: Bookmark & Organize Case Studies

- **Description:** Users can save case studies into custom research collections for reference, driving positive algorithmic signals.
- **Inputs:** Bookmark icon tap, collection assignment, folder creation name.
- **Outputs:** Active gold bookmark indicator, saved collection library, PWA offline cache sync, $5\times$ CAS telemetry event.
- **Functional Requirements:**
- **FR-04.1:** The system shall update bookmark state in $\le 50\text{ms}$ and dispatch a $5\times$ weighted telemetry boost to the author's CAS calculation pipeline.
- **FR-04.2:** The system shall allow users to organize saved cards into named custom folders.
- **FR-04.3:** The system's PWA service worker shall cache bookmarked slide text and images into IndexedDB for offline access.

3.2 Domain B: Content Authoring & Verification Layer

UC-05: Compose Case Study via Creation Studio

- **Description:** An optimized creation studio guiding consultants through a 4-step wizard (*Format $\to$ Template $\to$ Evidence $\to$ Conversion Anchor*) with split-screen preview.
- **Inputs:** Slide text, syntax-highlighted code snippets, Mermaid diagram scripts, media uploads.
- **Outputs:** Live interactive mobile preview pane, formatted multi-slide carousel deck.
- **Functional Requirements:**
- **FR-05.1:** The system shall provide 3 narrative frameworks: *Case Study* (*Challenge $\to$ Solution $\to$ Impact*), *Explainer* (*Concept $\to$ Anti-Pattern $\to$ Framework*), and *Toolkit* (*Problem $\to$ Stack $\to$ Checklist*).
- **FR-05.2:** The system shall enforce a strict limit of ≤ 300 characters per slide, disabling advancement if the limit is exceeded.

- **FR-05.3:** On desktop viewports, the system shall render real-time interactive slide updates on the right pane as text is entered on the left pane.

UC-06: Tag Evidence & Proof of Execution

- **Description:** Consultants attach verifiable proof artifacts (GitHub commits, enterprise client sign-offs, and masked case memos) to establish credibility.
- **Inputs:** GitHub repository URL, commit SHA, client username handle, image upload with redaction mask.
- **Outputs:** Verified GitHub repository badge, client co-author confirmation tag, irreversible redacted image assets.
- **Functional Requirements:**
- **FR-06.1:** The system shall query the GitHub API to authenticate repository existence, commit SHAs, star counts, and language breakdown.
- **FR-06.2:** The system shall notify tagged enterprise client accounts, displaying a "Client Verified" badge once confirmed.
- **FR-06.3:** The system shall provide an in-studio client-side canvas blur tool that permanently masks sensitive corporate IP before asset upload.

UC-07: Attach Conversion Anchor Call-to-Action

- **Description:** Consultants embed a native commercial conversion hook at the footer of their case studies.
- **Inputs:** Anchor type selection (*15-Min Discovery*, *Technical Audit*, *Download Framework*, *Retainer Inquiry*), pricing in KES, intake labels.
- **Outputs:** High-contrast sticky CTA button bound directly to the consultant's calendar or escrow vault.
- **Functional Requirements:**
- **FR-07.1:** The system shall support 4 distinct conversion anchor types with custom KES pricing.
- **FR-07.2:** The system shall scan CTA inputs and automatically reject any external payment links, WhatsApp numbers, or bypass URLs.
- **FR-07.3:** Clicking the CTA shall open the native slide-up booking sheet over the active content.

UC-08: Verify Identity & Professional Credentials

- **Description:** Automated vetting of consultant identity via government registries and statutory professional boards.
- **Inputs:** National ID number, legal full name, KRA PIN, statutory board practice number (LSK, ICPAK, EBK).
- **Outputs:** Bronze Verification badge (Identity/KRA valid), Silver Verification badge (Board active), CAS multiplier.
- **Functional Requirements:**
- **FR-08.1:** The system shall query IPRS and KRA gateways to confirm legal identity within $\le 60\text{s}$ to issue Bronze status.

- **FR-08.2:** The system shall query statutory professional board registries to confirm active practice standing, issuing Silver status and domain authority boosts.
- **FR-08.3:** If an annual credential re-verification fails, the system shall strip badges and place wallet withdrawals on provisional hold.

3.3 Domain C: Native Scheduling & Consultation

UC-09: Book In-App Advisory Consultation

- **Description:** Native slide-up scheduling sheet allowing clients to book advisory sessions in ≤ 4 taps without external redirection.
- **Inputs:** Slot selection, mandatory 3-question intake responses (*Challenge*, *Tech Stack*, *Expected Deliverable*).
- **Outputs:** 10-minute distributed mutex slot lock, calendar reservation record.
- **Functional Requirements:**
- **FR-09.1:** The system shall fetch real-time availability from Google and Outlook APIs, applying working hours and buffer constraints.
- **FR-09.2:** The system shall acquire an exclusive 10-minute Redis mutex lock on the selected slot upon proceeding to checkout.
- **FR-09.3:** The system shall mandate completion of all 3 intake questionnaire fields before unlocking payment rails.

UC-10: Process Payment via M-Pesa STK Push

- **Description:** Mobile checkout dispatching automated STK prompts directly to the client's Safaricom handset.
- **Inputs:** Safaricom mobile phone number, user PIN confirmation on mobile device.
- **Outputs:** STK push prompt, transaction webhook callback, confirmed booking record, eTIMS tax invoice receipt.
- **Functional Requirements:**
- **FR-10.1:** The system shall call Daraja / Paystack APIs to dispatch an STK push prompt within $\le 5\text{s}$ of submission.
- **FR-10.2:** The system shall confirm booking within $\le 2\text{s}$ of receiving a valid payment webhook, enforcing idempotency via `MpesaReceiptNumber`.
- **FR-10.3:** If the STK prompt times out after 60 seconds, the system shall render manual Paybill instructions with a 15-minute dynamic reference code.

UC-11: Process Cross-Border Card Payment

- **Description:** Multi-currency payment processing accepting Visa, Mastercard, and Amex from international enterprise clients.
- **Inputs:** Currency selection (USD, EUR, GBP, KES), credit/debit card details, 3D-Secure authentication.

- **Outputs:** Tokenized card record, FX conversion breakdown based on live CBK rates, confirmed booking.
- **Functional Requirements:**
- **FR-11.1:** The system shall fetch live Central Bank of Kenya reference exchange rates and display KES equivalents before checkout.
- **FR-11.2:** The system shall mandate 3D-Secure 2.0 two-factor authentication for all card transactions.
- **FR-11.3:** The system shall tokenize card credentials in compliance with PCI-DSS, storing only masked PAN digits and secure tokens.

UC-12: Conduct Secure Live Consultation

- **Description:** Confidential advisory sessions conducted in-browser via encrypted WebRTC video.
- **Inputs:** Camera/microphone streams, screen share capture, sidebar action-item text entries.
- **Outputs:** Peer-to-peer 1080p video, 5-minute wrap-up reminder banner, emailed session summary.
- **Functional Requirements:**
- **FR-12.1:** The system shall establish encrypted peer-to-peer WebRTC connections via DTLS-SRTP, falling back to TURN relay servers when direct connection is blocked.
- **FR-12.2:** The system shall support 1080p 30fps screen sharing with interactive diagram markup tools.
- **FR-12.3:** The system shall display a non-intrusive banner when 5 minutes remain in the session, allowing mutual 15-minute extensions.

3.4 Domain D: Milestone Escrow & B2B Contracts

UC-13: Fund Milestone Escrow Contract

- **Description:** De-risks multi-week consulting engagements by locking project funds into an escrow vault prior to execution.
- **Inputs:** Itemized Statement of Work (SOW) milestone schedule, digital signatures, escrow deposit payment.
- **Outputs:** Immutable escrow contract record, locked deposit balance, execution authorization notice.
- **Functional Requirements:**
- **FR-13.1:** The system shall require both parties to digitally sign the SOW defining deliverables, deadlines, and milestone budgets.
- **FR-13.2:** The system shall lock deposit funds into the platform trust ledger upon payment clearance.
- **FR-13.3:** The system shall prevent the consultant from advancing to milestone execution until escrow funding is verified.

UC-14: Submit Project Milestone Deliverable

- **Description:** Consultants upload final deliverables, generating immutable cryptographic delivery proofs.
- **Inputs:** Deliverable asset files (PDFs, ZIPs, code repositories), technical completion summary, extension requests.
- **Outputs:** Uploaded locker assets, SHA-256 checksum recorded in ledger, 72-hour review timer initialized.
- **Functional Requirements:**
- **FR-14.1:** The system shall compute a SHA-256 checksum across all uploaded files and record the hash and timestamp in the immutable audit ledger.
- **FR-14.2:** The system shall initialize a 72-hour inspection countdown timer upon deliverable submission.
- **FR-14.3:** The system shall allow consultants to request deadline extensions without penalty if submitted before the milestone due date.

UC-15: Review & Approve Escrow Milestone

- **Description:** Client reviews deliverables, releasing funds with 1 click; includes automated 7-day auto-release fallbacks and instant mobile payouts.
- **Inputs:** Client "Approve Deliverable" click, withdrawal destination selection (M-Pesa B2C vs Bank EFT).
- **Outputs:** Unlocked escrow funds, 5%–8% platform take rate journalized, net payout credited, instant B2C mobile transfer.
- **Functional Requirements:**
- **FR-15.1:** Upon client approval, the system shall release funds, deduct the platform take rate, and credit the consultant wallet in $\le 5\text{s}$.
- **FR-15.2:** If a client takes zero action for 168 hours (7 days) after submission, the system shall automatically execute the approval and release funds.
- **FR-15.3:** The system shall disburse funds to the consultant's Safaricom phone via B2C API in under 30 seconds upon withdrawal request.

UC-16: Initiate Escrow Dispute & Mediation

- **Description:** Halts automated release clocks when deliverables fail specifications, opening an evidence-backed dispute process.
- **Inputs:** Client/consultant dispute notice, SOW clause breach descriptions, uploaded evidence files.
- **Outputs:** Frozen escrow balance, paused countdown timers, mutual 48-hour evidence submission thread.
- **Functional Requirements:**
- **FR-16.1:** Raising a dispute shall immediately freeze milestone escrow funds and pause auto-approval timers.

- **FR-16.2:** The system shall provide a 48-hour mutual evidence submission portal for both parties.
- **FR-16.3:** The system shall allow parties to execute an amicable prorated split agreement without mediator intervention.

3.5 Domain E: Algorithmic Intelligence & Telemetry

UC-17: Calculate Creator Authority Score (CAS)

- **Description:** Evaluates professional authority using a multi-factor formula, replacing superficial vanity metrics.
- **Inputs:** Credential records, completed escrow contracts, star ratings, save/bookmark telemetry, peer endorsements.
- **Outputs:** Normalized CAS score (0.00 to 10.00), algorithmic feed ranking multiplier.
- **Functional Requirements:**
- **FR-17.1:** The system shall calculate CAS based on: Verified Credentials (40%), Completed Contracts & Ratings (30%), Sanitized Bookmark Telemetry (15%), and Peer Endorsements (15%).
- **FR-17.2:** The system shall apply a 5% monthly decay to engagement factors after 180 days of creator inactivity.
- **FR-17.3:** The system shall detect and quarantine bookmark velocity anomalies originating from bot farms or correlated IP addresses.

UC-18: Personalize Two-Pass Discovery Feed

- **Description:** Two-pass recommendation pipeline delivering hyper-relevant feeds with regional East African context.
- **Inputs:** User interaction vectors, 128-D case study vectors, creator CAS scores, geographic tags.
- **Outputs:** Ranked top 20 feed cards delivered in under 200ms.
- **Functional Requirements:**
- **FR-18.1:** **Pass 1:** The system shall retrieve the top 200 candidate cards using vector cosine similarity within $\le 50\text{ms}$.
- **FR-18.2:** **Pass 2:** The system shall score candidates by multiplying vector similarity $\times$ CAS $\times$ Regional East Africa Multiplier ($1.25\times$), returning top 20 cards in $\le 200\text{ms}$.
- **FR-18.3:** The system shall inject a 15% exploration quota across diverse disciplines for newly onboarded users.

3.6 Domain F: Governance, Administration & Compliance

UC-19: Manage Phased Rollout & Creator Onboarding

- **Description:** Manages ecosystem transitions across Phase 1 (Anchor Agency), Phase 2 (Curated Cohort), and Phase 3 (Open Multi-Tenancy).
- **Inputs:** Admin phase state toggle, cryptographic invite code issuance, applicant registration submissions.
- **Outputs:** Studio access permission gates, Founding Creator badges, multi-tenant directory indexing.
- **Functional Requirements:**
- **FR-19.1:** In Phase 1, the system shall restrict publishing exclusively to proprietary founding agency accounts.
- **FR-19.2:** In Phase 2, creator registration shall require a valid cryptographic invitation code.
- **FR-19.3:** In Phase 3, the system shall unlock automated self-service onboarding with multi-tenant workspace isolation.

UC-20: Moderate Platform Content & Enforce NDAs

- **Description:** NLP quality filtering and emergency takedowns to protect enterprise intellectual property.
- **Inputs:** Published posts, community flags, enterprise NDA takedown requests, creator appeals.
- **Outputs:** Quarantined post records, emergency 4-hour SLA admin tickets, moderation rulings.
- **Functional Requirements:**
- **FR-20.1:** The system shall execute automated NLP scans to quarantine posts containing spam, emotional fluff, or lacking technical frameworks.
- **FR-20.2:** Verified enterprise NDA complaints shall immediately hide the post within $\le 60\text{s}$ and trigger a 4-hour SLA admin review ticket.
- **FR-20.3:** The system shall provide an appeal portal allowing creators to submit justifications for flagged posts.

UC-21: Adjudicate Milestone Dispute

- **Description:** Neutral dispute resolution dashboard enabling certified mediators to review evidence and issue binding rulings.
- **Inputs:** SOW contracts, SHA-256 asset checksums, chat transcripts, mediator percentage ruling.
- **Outputs:** Executed fund disbursements, immutable adjudication log entry, CIArb arbitration referral.
- **Functional Requirements:**
- **FR-21.1:** The system shall display all SOW clauses, SHA-256 asset checksums, and communication logs in a unified chronological dashboard.
- **FR-21.2:** The system shall execute binding percentage disbursements (0%–100%) as directed by the mediator's authenticated ruling.

- **FR-21.3:** For contracts $> \text{KES } 1,000,000$, the system shall support escrow transfer into legal suspense pending formal CIArb tribunal awards.

UC-22: Audit Financial Settlement & Tax Compliance

- **Description:** Automated revenue accounting, KRA eTIMS fiscal invoice generation, and withholding tax compliance.
- **Inputs:** Milestone release events, consultation payments, corporate tax PINs.
- **Outputs:** 5%–8% platform take rate journal entries, KRA eTIMS QR-coded VAT receipts, 5% WHT certificates.
- **Functional Requirements:**
- **FR-22.1:** The system shall automatically deduct the 5%–8% platform take rate and credit it to the platform operating account upon payment release.
- **FR-22.2:** The system shall call the KRA eTIMS API to generate a cryptographically validated tax invoice with a scannable QR code for every fee transaction.
- **FR-22.3:** The system shall itemize statutory 5% professional Withholding Tax (WHT) deductions on corporate billing statements.

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4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces & Dual-Mode Ergonomics

- The UI shall adhere to a dual-mode interaction model:

13. **Vertical Stream (Home Tab)**: Full-height, swipe-driven feed for passive discovery.
14. **Bento Grid (Profile & Explore Tabs)**: High-density modular grid for 5-second competence due diligence.

- The UI shall support both light and dark modes with high-contrast typography satisfying WCAG 2.1 AA ($\ge 4.5:1$).

4.2 Hardware Interfaces & Camera/Sensor Streaming

- The system shall access local video capture hardware via the HTML5 `MediaDevices.getUserMedia()` interface.
- The system shall process touch events from mobile capacitive touchscreens, distinguishing between horizontal slide drags ($>15\text{px}$) and vertical feed swipes.

4.3 Software Interfaces & Statutory Registries

- **Government Identity (IPRS & KRA)**: Integration via REST/JSON to validate National ID numbers and KRA PIN certificates.
- **Professional Statutory Boards**: Connectors to Law Society of Kenya (LSK), ICPAK, and Engineers Board of Kenya (EBK) to verify active license standing.

- **Developer Ecosystems**: GitHub REST API v3 integration to authenticate repositories, pull commit history, and verify code authorship.

4.4 Communications & Fintech Protocol Interfaces

- **Safaricom Daraja API**:
- STK Push (C2B): `POST /mpesa/stkpush/v1/processrequest` over TLS 1.3.
- Callback Webhook: Secure HTTPS webhook receiver with IP whitelisting and HMAC signature verification.
- B2C Mobile Payouts: `POST /mpesa/b2c/v1/paymentrequest`.
- **Card Rails (Paystack / Flutterwave)**: PCI-DSS Level 1 compliant tokenized iframe with 3D-Secure 2.0 challenge authentication.
- **KRA eTIMS Fiscal Gateway**: Mutual TLS (mTLS) REST API for electronic tax invoice signing and fiscal QR generation.

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5. NON-FUNCTIONAL QUALITY REQUIREMENTS

```
mindmap
  root((ISO/IEC 25010 Quality Standards))
    Performance
      P95 Feed Latency <= 200ms
      Vector Search <= 50ms
      App Shell TTI <= 1.2s
    Security & Privacy
      PostgreSQL RLS 100%
      Kenya DPA 2019 AES-256
      WebRTC DTLS-SRTP
    Reliability & Fault Tolerance
      Escrow Zero Balance Drift
      STK Webhook Idempotency
      Circuit Breaker <= 15s
    Usability & Governance
      5-Second Due Diligence Scan
      Zero-Redirect Native Booking
      4-Hour SLA NDA Takedown
```

5.1 Performance Requirements

- **NFR-PERF-01**: Vertical feed card transition latency (P_{95}) shall not exceed 200ms on 4G LTE connections.
- **NFR-PERF-02**: Pass 1 vector search retrieval across 100,000 indexed case studies shall execute within $\le 50\text{ms}$.
- **NFR-PERF-03**: Time to Interactive (TTI) for the application shell shall not exceed 1.2 seconds on mid-tier mobile hardware.

5.2 Security, Confidentiality & RLS Requirements

- **NFR-SEC-01**: The database shall enforce Row-Level Security (RLS) on all tenant-sensitive tables, with automated CI tests asserting zero unauthorized cross-tenant data leaks.
- **NFR-SEC-02**: All user PII shall be encrypted at rest using AES-256 and in transit using TLS 1.3, satisfying the Kenya Data Protection Act (DPA 2019).
- **NFR-SEC-03**: All WebRTC video media streams shall be end-to-end encrypted via DTLS-SRTP.

5.3 Reliability, Availability & Escrow Consistency

- **NFR-REL-01**: The escrow double-entry ledger shall maintain 100% ACID consistency with zero balance drift ($\sum \text{Deposits} = \sum \text{Held Escrow} + \sum \text{Disbursements} + \sum \text{Fees}$).
- **NFR-REL-02**: The M-Pesa webhook handler shall enforce 100% idempotency, safely ignoring duplicate receipt callbacks.
- **NFR-REL-03**: Upstream payment gateway failures shall trip a circuit breaker within 15 seconds, seamlessly activating fallback payment rails.

5.4 Usability & Accessibility (WCAG 2.1 AA)

- **NFR-USA-01**: A client visiting a consultant's profile shall be able to identify their domain authority, rating, and primary discipline within $\le 5\text{ seconds}$ above the fold.
- **NFR-USA-02**: Advisory booking shall complete in ≤ 4 distinct user taps within an in-app slide-up sheet without external redirects.
- **NFR-USA-03**: All user interface elements shall satisfy WCAG 2.1 Level AA color contrast requirements ($\ge 4.5:1$).

5.5 Maintainability, Extensibility & Modifiability

- **NFR-MNT-01**: Transitions across rollout phases (Phase 1 Agency $\to$ Phase 2 Beta $\to$ Phase 3 Open Ecosystem) shall be executed via configuration flags with zero database migrations or downtime.
- **NFR-MNT-02**: Reported enterprise NDA breaches shall be automatically quarantined within $\le 60\text{s}$ and reviewed under a strict 4-hour SLA.

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6. VERIFICATION MATRIX & ACCEPTANCE CRITERIA

Feature / Use Case	Applicable Quality Requirements	Verification Method	Acceptance Criteria
UC-01 (Dual-Mode Feeds)	`NFR-PERF-01`, `NFR-PERF-03`	Automated Telemetry & Lighthouse	P_{95} transition $\le 200\text{ms}$; video auto-downgrades to WebP on $<1\text{ Mbps}$.
**UC-02 (Bento Grid	`NFR-USA-01`, `NFR-SEC-	Usability Testing & Pen-	Authority baseline visible

Profile)**	01`	Test	in $\le 5\text{s}$; zero cross-tenant profile data leak.
UC-03 (Topic Hubs Search)	`NFR-PERF-02`	Load Testing (k6)	200-candidate vector search $\le 50\text{ms}$ at 100k corpus size.
UC-04 (Bookmarks)	`NFR-PERF-01`	Unit & Integration Tests	Tap updates state in $\le 50\text{ms}$; dispatches $5\times$ telemetry weight to CAS engine.
UC-05 (Creation Studio)	`DC-03`, `NFR-USA-03`	Automated Form Testing	Character limit enforces hard block at 301 chars; split preview syncs in real time.
UC-06 (Evidence Tagging)	`NFR-SEC-02`	API Mock & Security Audit	GitHub commit authenticated; redacted images permanently mask underlying bytes.
UC-07 (Conversion Anchor)	`DC-04`	Regex & Security Scanner	Phone numbers/external payment URLs rejected; binds natively to calendar.
UC-08 (Credential Verify)	`NFR-SEC-02`	Gateway Integration Tests	IPRS/KRA validates in $\le 60\text{s}$; invalid license triggers provisional hold.
UC-09 (Native Booking)	`DC-04`, `NFR-USA-02`	Concurrency Testing	Slide-up modal completes in ≤ 4 taps; slot locked with 10-min Redis mutex.
UC-10 (M-Pesa STK Push)	`NFR-REL-02`, `NFR-REL-03`	Chaos & Idempotency Testing	STK dispatched $\le 5\text{s}$; duplicate webhooks handled idempotently without error.
UC-11 (Card Payments)	`NFR-SEC-02`	PCI-DSS Compliance Scan	Live CBK FX conversion displayed; 3D-Secure 2.0 challenge required.
UC-12 (Live Consultation)	`NFR-SEC-03`	Packet Inspection & WebRTC	DTLS-SRTP end-to-end encryption confirmed; 5-min overtime banner displays.

UC-13 (Fund Escrow)	`NFR-REL-01`	ACID Ledger Concurrency Test	Milestone locked in escrow ledger; zero balance drift across transactions.
UC-14 (Deliverable Proof)	`NFR-SEC-02`	Checksum Verification	Uploaded files stamped with SHA-256 hash and ISO timestamp in ledger.
UC-15 (Approve & Payout)	`NFR-REL-01`	Automated Billing Audit	Funds unlock in $\le 5\text{s}$; 7-day auto-release executes if client is inactive.
UC-16 (Escrow Dispute)	`NFR-REL-01`	Integration Test	Dispute pauses all timers; 48-hour mutual evidence portal initializes.
UC-17 (Calculate CAS)	`NFR-MNT-01`	Mathematical Unit Test	40/30/15/15 formula evaluated deterministically; 180-day decay applied.
UC-18 (Two-Pass Feed AI)	`NFR-PERF-01`, `NFR-PERF-02`	Synthetic End-to-End Test	Pass 1 vector $\le 50\text{ms}$; Pass 2 heavy ranking delivers top 20 in $\le 200\text{ms}$.
UC-19 (Phased Rollout)	`NFR-MNT-01`	Deployment State Test	Phase transitions execute via config flags with zero schema migration downtime.
UC-20 (Content Moderation)	`NFR-MNT-02`	SLA Drill Testing	Spam quarantined; enterprise NDA reports hidden in $\le 60\text{s}$ with 4h review SLA.
UC-21 (Adjudicate Dispute)	`NFR-REL-01`	Simulation Audit	Mediator executes binding split; contracts $> \text{KES } 1\text{M}$ offer CIArb referral.
UC-22 (Financial Tax Audit)	`DC-02`, `NFR-REL-01`	Tax Gateway Validation	5-8% take rate journalized; KRA eTIMS QR invoice generated for all platform fees.

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7. DOCUMENT APPROVAL & SIGN-OFF